

MA540 Project 6

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Statement

We consider the Dittus-Boelter equation

$$Nu = \theta_1 Re^{\theta_2} Pr^{\theta_3}$$

where Nu , Re and Pr respectively denote the Nusselt, Reynolds, and Prandtl numbers. We will use the reported nominal values

$$\theta^0 = [0.023, 0.8, 0.4]$$

and data provided in the file `db_data.txt`, where `Re = db_data(:,1); Pr = db_data(:,2); Nu = db_data(:,3)`. We will construct the Fisher information matrix and discuss the identifiability of the parameters. We will then use DRAM to compute posterior densities for the parameters, and compare the pairwise plots to the Fisher information results.

Solution

We use `fminsearch` to optimize the parameters with OLS, using the given nominal values. We report the optimized values $\hat{\theta} = [0.0040, 0.9863, 0.4108]$. We then use complex-step approximation to compute sensitivities for each parameter, using $h = 10^{-20}$. We then construct the sensitivity matrix X , and compute $\sigma^2 = \frac{1}{N-p} \sum R^2$, where R are the residuals between Nu values from `db_data.txt` and Nu values calculated with $\hat{\theta}$, with $N = 56$ and $p = 3$. We compute the Fisher Information matrix, obtaining:

$$X^T X = 10^{10} \times \begin{array}{|c|c|c|} \hline 9.3925 & 0.3577 & 0.1331 \\ \hline 0.3577 & 0.0137 & 0.0050 \\ \hline 0.1331 & 0.0050 & 0.0021 \\ \hline \end{array}$$

We compute $\text{Rank}(X^T X) = 3$, which indicates that all three parameters are identifiable. We compute eigenvalues $\lambda_1 = 2.932 \times 10^5$, $\lambda_2 = 2.724 \times 10^6$, $\lambda_3 = 9.408 \times 10^{10}$. None of these eigenvalues are less than a reasonable threshold ϵ , so we confirm that our local sensitivity analysis finds that all parameters are locally identifiable.

We now run DRAM, initialized with the optimized parameters, the covariance matrix $V = \sigma^2(X^T X)^{-1}$, and variance σ^2 . We run $N = 10^5$ iterations. Figure 1 shows the chains for each parameter under DRAM. We see each exhibiting "fuzzy caterpillar" behavior, indicating high exploration and minimal stagnation.

Figure 1: DRAM Chains

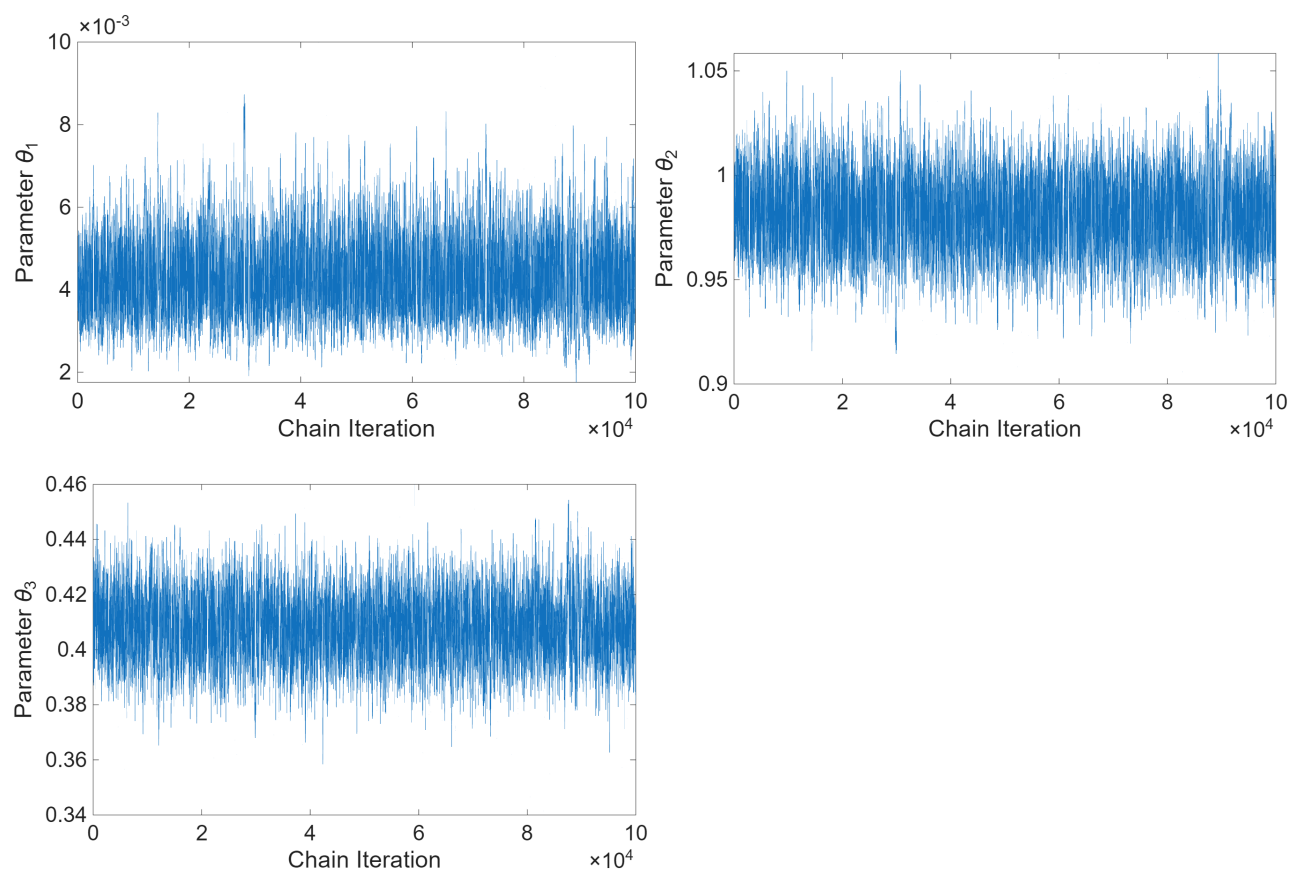
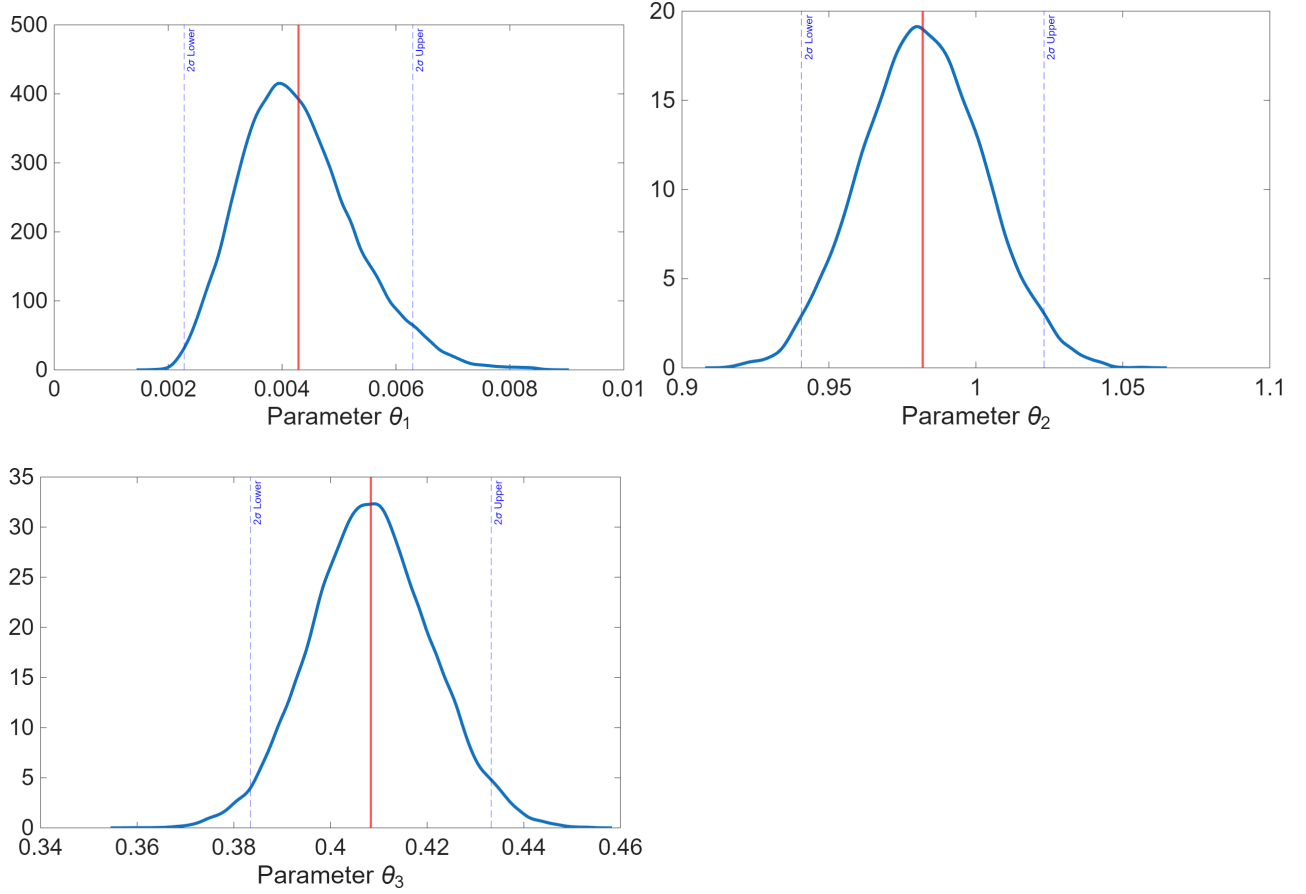


Figure 2: Posterior Densities from DRAM



We then use `kde.m` to construct posterior densities for the parameters, using the DRAM chains. Figure 2 shows the posterior densities for each parameter, along with 2σ intervals.

Figure 3 shows the pairwise plots for each parameter pair. θ_3 shows some linear posterior correlation with θ_1 and θ_2 , while θ_1 and θ_2 exhibit strong nonlinear posterior correlation. This indicates that globally, θ_1 and θ_2 are not individually identifiable. Hence, the pairwise plots are not consistent with the Fisher information results. The Fisher information results are local, while the DRAM results are global, so we posit that the parameters are locally identifiable around the optimized $\hat{\theta}$ values, but globally, θ_1 and θ_2 are not individually identifiable.

Figure 3: Pairwise Plots from DRAM

